LHCb XY surface setup: lhcb_lower_up_y8_ft_10_10_12

Selected reusable detector planes for transport and acceptance masks.

detector	surfaces	fired layers	z range [mm]	XY envelope [mm]	description
TV	50	50	-287.5 to 557.6	90.3 x 92.0	VELO/TV layers formed by unioning the nearby-z sensor faces in each layer.
RICH1	2	2	795.0 to 2237.2	1384.1 x 1142.5	Projective RICH1 radiator entry/exit surfaces from angular acceptance.
UP	16	4	2314.0 to 2656.0	1393.7 x 964.9	Upstream Tracker surfaces; choose nominal GDML or Yisheng-inspired border crops.
MP	5	5	7665.5 to 9132.5	3288.4 x 672.6	Five Mighty Pixel layers, each formed from the union of its extracted tiles.
FT	12	12	7902.7 to 9384.7	6706.9 x 4891.0	Mighty Tracker X,U,V,X stations; choose 8/10/12 module widths per T station.
RICH2	2	2	9487.5 to 11839.2	2855.1 x 2375.8	Projective RICH2 radiator entry/exit surfaces from angular acceptance.
PicoCal	2	2	12565.9 to 12601.0	7801.6 x 6338.8	Two front-region PicoCal surfaces, entry and exit, as a simple volume proxy.
MUON	4	4	15270.0 to 18870.0	12119.2 x 10057.3	Muon stations M2-M5, rectangular acceptances with central beam holes.

Configuration

{"ft": {"stations": {"T1": 10, "T2": 10, "T3": 12}}, "include": ["TV", "RICH1", "UP", "MP", "FT", "RICH2", "PicoCal", "MUON"], "name": "lhcb_lower_up_y8_ft_10_10_12", "source": "ViewLHCb lower ROOT/GDML extracted surfaces", "tv": {"grouping": "paire